1) October 12 Classwork: Runner.java (code and output) - 5 points

```java
import java.util.Scanner;

public class Runner{



   public static void main(String[] args) {

       int count = 1;

       while(count<13){

           System.out.print(count + " ");

           count++;

       //counts up to 12 - increment 1

       }

       System.out.println();

       count = 0;

       while(count <= 18){

           System.out.print(count + " ");

           count += 2;

             //counts up to 18 - increment 2



       }

       System.out.println();

       count = 15;

       while(count >= -10){

           System.out.print(count + " ");

           count -= 5;



       }
```

```
        System.out.println();



        Scanner sc = new Scanner(System.in);




        int startnum = 0;



        //Asks the user what number to count up by, and also what increment to go by

        System.out.print("Provide a number that is greater than 20: ");

        int endnum = sc.nextInt();

        System.out.println("Provide a step (What will be counted by) ");

        int step = sc.nextInt();



        while(startnum <= endnum ){

            System.out.print(startnum + " ");

            startnum += step;



        }

        System.out.println();





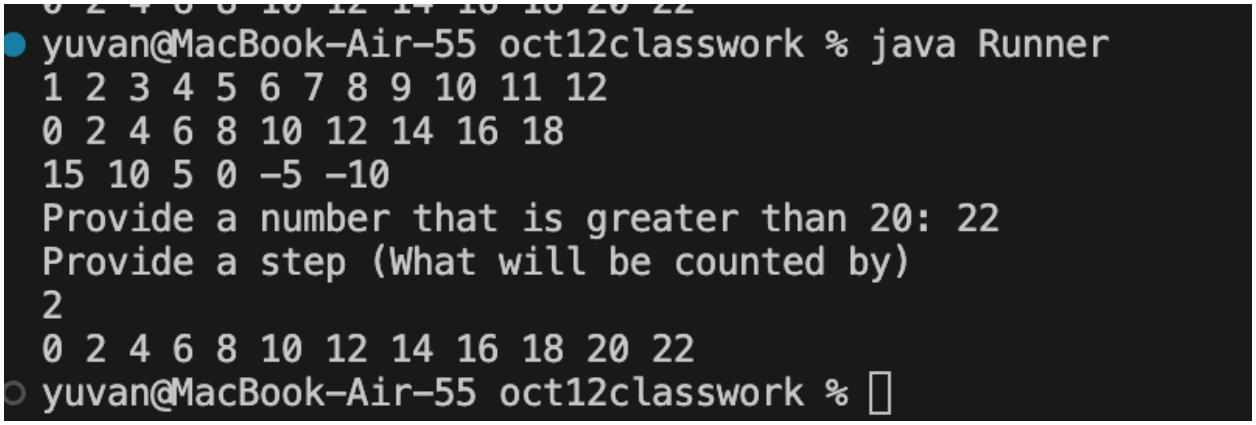




    }
```

```
}
```

Output:



2) October 13/14 Warm-Up: Runner.java and Pythagorean.java (code and output) - 5 points

Runner:

```
public class Runner {

   public static void main(String[] args){



      //instantiates obj as a object of the pyth class, then calls find c with the
sides set to 12 and 5

      //then it prints the result



      Pythagorean obj1 = new Pythagorean();
```

```
        double result = obj1.findC(12,5);

        System.out.println(result);




    }







}
```

Pythagorean:

```
public class Pythagorean {

    private double a;

    private double b;

    private double c;



    Pythagorean(){

        //Default Contructor



    }



    public double findC(double a, double b){



        //a^2 + b^2 = C^2

        c = Math.sqrt(Math.pow(a, 2) + Math.pow(b,2));

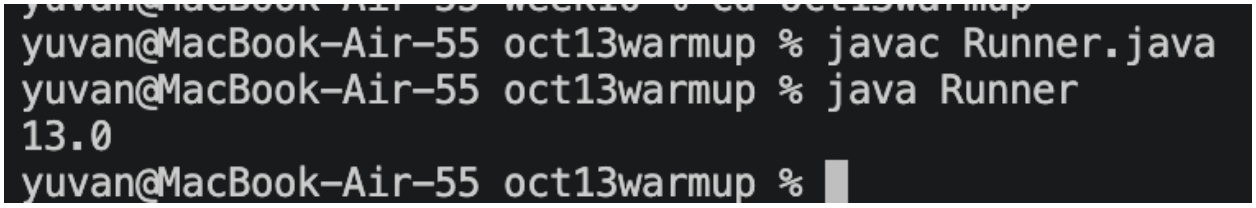
        //Calculates for c variable in the pythagorean theorem
```

```
        return c;


    }


}
```

Output:



3) October 13/14 Classwork: Runner.java (code and output) - 5 points

```
import java.util.Scanner;

public class Runner {



   public static void main(String[] args) {

        //while loop that adds up all numbers between 2 and 19 inclusive (should be
189)

        int count = 2;

        int sum = 0;



        while (count <= 19) {

        sum += count;

        count++;

        }

        System.out.println(sum);
```

```
      count = 1;
      sum = 0;

      do{
          count ++;
          sum += count;


      }while(count < 19);
      System.out.println(sum);

      Scanner sc = new Scanner(System.in);


      System.out.print("Provide a positive number: ");
      int input = sc.nextInt();
      //checks whether the input is less than or is zero, and if that is true, it
rejects the input
      //and reprompts the user.
      while (input <= 0) {
      System.out.print("That's not a number. Provide a positive number: ");
      input = sc.nextInt();
      }

      int startnum = 0;
```

```
        do {
            startnum ++;
            System.out.print(startnum + " ");


        }while(startnum < input);

        System.out.println();
        startnum = input;
        do{
            System.out.print(startnum + " ");
            startnum--;
        }while(startnum >= 1);

        System.out.println();
    }

}
```

Output:

```
yuvan@MacBook-Air-55 oct13classwork % java Runner
189
189
Provide a positive number: -1
That's not a number. Provide a positive number: -1
That's not a number. Provide a positive number: 5
1 2 3 4 5
5 4 3 2 1
```

4) October 13/14 while do while loop activity: WhichLoop.java - 5 points

```
import java.util.Scanner;
public class WhichWhile {

   public static void main(String[] args){
       Scanner sc = new Scanner(System.in);

       System.out.print("Input a number: ");

       int storednum = sc.nextInt();
       //when the number is less than or equal to 5, reject it and reask the user for
a new number.
       while (storednum <= 5){
           System.out.print("Input a number: ");
           storednum = sc.nextInt();
       }
       System.out.println("The number is: " + storednum);
   }

}
```

Output:

```
yuvan@MacBook-Air-55 oct13classwork2 % java WhichWhile
Input a number: 3
Input a number: 7
The number is: 7
yuvan@MacBook-Air-55 oct13classwork2 %
```

5) October 13/14 while do while loop activity: Fill in this sentence: - 5 points

When deciding whether to use a while or do-while loop, you must consider when you want the condition to be checked, as this is the main difference between these two loops. You can either have the condition checked at the start or at the end.

6. October 15/16 Warm-Up: LoopError.java (code and output), and answers to the two questions

Code:

```
public class LoopError {




   public static void main(String[] args) {




      // generate a random number between 5 and 25 inclusive

      int rand = (int)(Math.random()*21) + 5;

      System.out.println(rand);

      // Use a do while loop to add up all of the numbers from 1 to the random
number, inclusive

      System.out.print("The sum of the numbers from 1 to " + rand + " are ");

      //loop logic that calculates the addition

      int sum;

      sum = 0;

      do {



          sum += rand;

          rand--;
```

```
        } while (rand >= 1); //condition checked



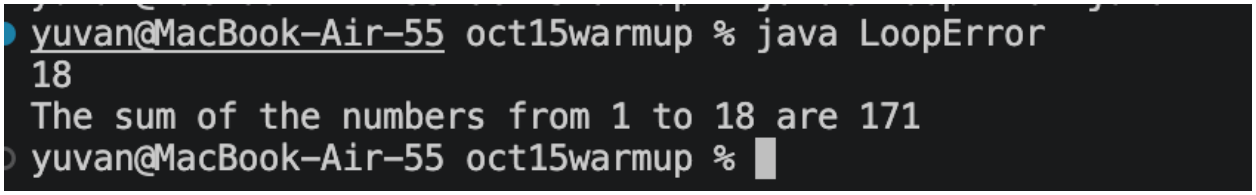
        //prints out calculated value (sum)

        System.out.println(sum);

    }

}
```

Output:



Questions:

a.)What caused the code to not work as intended?

The code did not work as intended because, in the do code block, sum was reset to zero every time. This prevented the numbers from adding up because the variable tracking the sum was reset to zero. The fix is to put the line of code that sets sum = 0 outside of the do code block (above it)

b.)What type of error was this (runtime, semantic, syntax), and why do you think that?

This error is semantic because the code does not run as the user intended. There is also no runtime error or compiler error that shows up in the terminal after running this code

7. October 15/16 Classwork: BooLean.java (code and output) - 10 points

```
import java.util.Scanner;



public class BooLean {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
```

```
int points = 20;



while (true) {

    // pick two random numbers from 1 to 10

    int num1 = (int)(Math.random() * 10 + 1);

    int num2 = (int)(Math.random() * 10 + 1);

    System.out.println(num1 + " " + num2);



    if (num1 == 5 && num2 == 5) {

        System.out.println("you win 10 points");

        points += 10;

    } else if (num1 == num2) {

        System.out.println("you win 3 points");

        points += 3;

    } else if (num1 % 2 == 0 || num2 % 2 == 0) {

        System.out.println("you win 2 points");

        points += 2;

    } else if (num1 > 3 || num2 > 3) {

        System.out.println("you win 1 point");

        points += 1;

    } else {

        System.out.println("you lose 1 point");

        points -= 1;

    }

    System.out.println("Points: " + points);



    // stop the game if the user doesn't type y
```

```
            System.out.println("Play again? (y/n)");

            if (!input.nextLine().equals("y")) {

                break;

            }

        }

    }

}
```

Output:

```
yuvan@MacBook-Air-55 oct15classwork % java BooLean
8 3
you win 2 points
Points: 22
Play again? (y/n)
y
5 5
you win 10 points
Points: 32
Play again? (y/n)
n
yuvan@MacBook-Air-55 oct15classwork %
```